LAB-4 REPORT

ABSTRACT: -

Initially a driver was written for the collection of data from the IMU and GPS. A total of 2 datasets were collected. One dataset was collected while moving in the circular pattern which helped for Magnetometer calibration while the other dataset was collected while driving across the city and on this data the further analysis was done.

The driver for LAB4 and the ROSBAGS are posted by "Divyansh Gupta" on his GITLAB account with username "gupta.divy".

REPORT STRUCTURE

PART 1: -

- Magnetometer Calibration
- Yaw Calculation using Magnetometer and gyro

PART 2: -

- Velocity estimation from accelerometer and comparing that with GPS velocity

PART 3: -

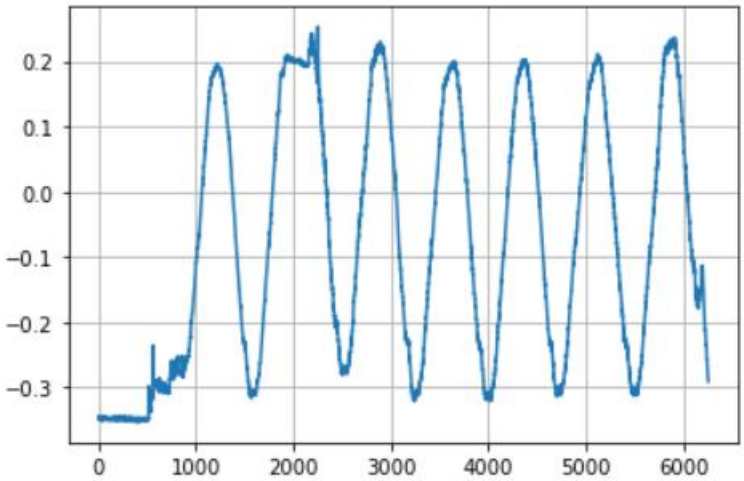
- Total distance travelled calculated from GPS and Accelerometer
- (Linear velocity in X) *(Angular Velocity in Z) vs Linear Acceleration in Y
- Trajectory estimation from IMU and comparing with trajectory given by GPS

PART 1: -

a.) Magnetometer Calibration.

The data collected from IMU while moving in circular trajectory is used to find the hard and soft Iron calibration values for the IMU.

As initially the magnetometer data has noise the range of data from Magnetometer X and Y was selected for which the magnitude of Magnetic field was constant and on this separated data the calibration values were found out.



Length of dataset vs Magnetometer_X values

The range of data on which the hard and soft iron calibration values are found is 3242 to 5528.

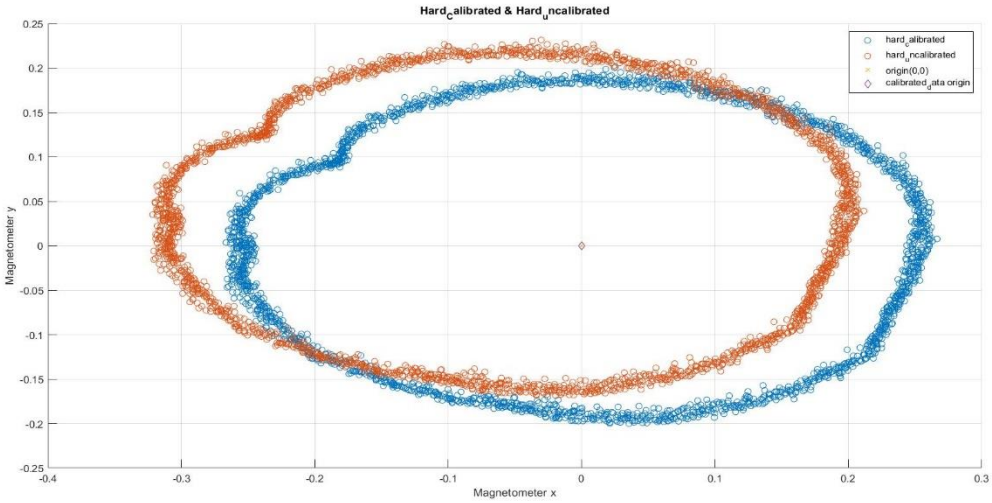
Hard Iron: - Account for translation along the origin in X and Y direction

Soft Iron: - Soft Iron distort the circle (As we are moving in circular path) and rotate it.

HARD IRON CALIBRATION

To consider the hard iron error the mean of range of data selected in both magnetometer X direction and magnetometer Y direction is taken separately and these are named as X offset and Y offset and these offset values account for the hard iron errors in our data. So simply subtracting these offsets from the corresponding Magnetometer X data and Magnetometer Y data removes hard iron errors.
X Offset = -0.0550
Y Offset = 0.0315

Once the hard iron error is removed from the data moving in circular path, soft iron calibration on the range of data specified above is done.



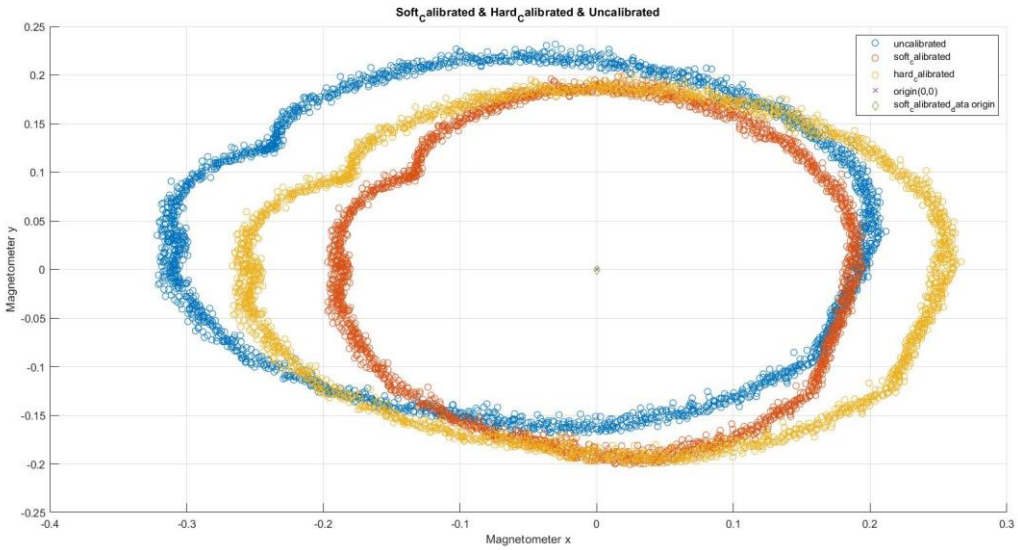
SOFT IRON CALIBRATION

First an ellipse was fitted to the selected range of data to account for the distortion of the circle and make it easy to rotate it back to ~0degree with respect to the horizontal X axis of the graph in which the data was plotted. The ellipse fitting was done using least square error criterion. With the help of fit_ellipse function in matlab the process was done. This function also gives the length of major axis and minor axis as well as rotation about a fixed frame in radians.

Length of Major Axis = 0.5118
Length of Minor Axis = 0.3779
Rotation with respect to X = -0.0144(radians) or -0.8259(degrees)

First the whole hard iron calibrated Magnetometer X and Magnetometer Y datasets are rotated through -0.0144(radians) using the rotation matrix formed and then the scale is calculated as length of Major axis divided by length of minor axis which gives a scale value of 1.3544. The scale value is then divided by hard iron calibrated magnetometer X data hence removing soft iron errors as well as hard iron errors from our data

Note :- Soft Iron calibration is done on hard iron calibrated data.



b.) Yaw Calculation using Magnetometer and Gyro

From this part until the end of the report the driving dataset is used.

The magnetometer X and Y data in the driving IMU dataset is first calibrated using the hard and soft iron calibration constants that we got from circular moving dataset.

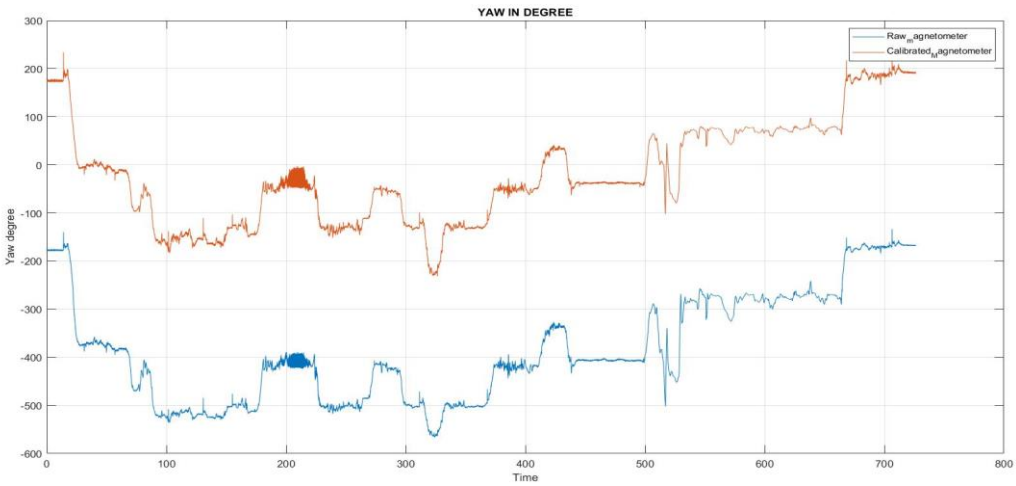
The X offset and Y offset is applied to driving Magnetometer X data and Magnetometer Y data respectively to get hard iron calibrated magnetometer data and then this data is rotated by -0.0144(radians) and the Magnetometer X data was scaled down by a factor of 1.3544 to get hard and soft iron calibrated data.

The Yaw from magnetometer is calculated by using atan2(-Mag_Y / Mag_X)

Tan2 instead of simply tan is used as tan2 takes care of the quadrant in which the Magnetometer Y and Magnetometer X data is and thus giving us Yaw correctly.

This formula is applied on both calibrated magnetometer data as well as uncalibrated magnetometer data. After getting yaw this way in radians, the yaw angles(in radians) is unwrapped in both the cases as atan2 bounds the angle between -180 to 180. Then after unwrapping the yaw is converted to degrees.

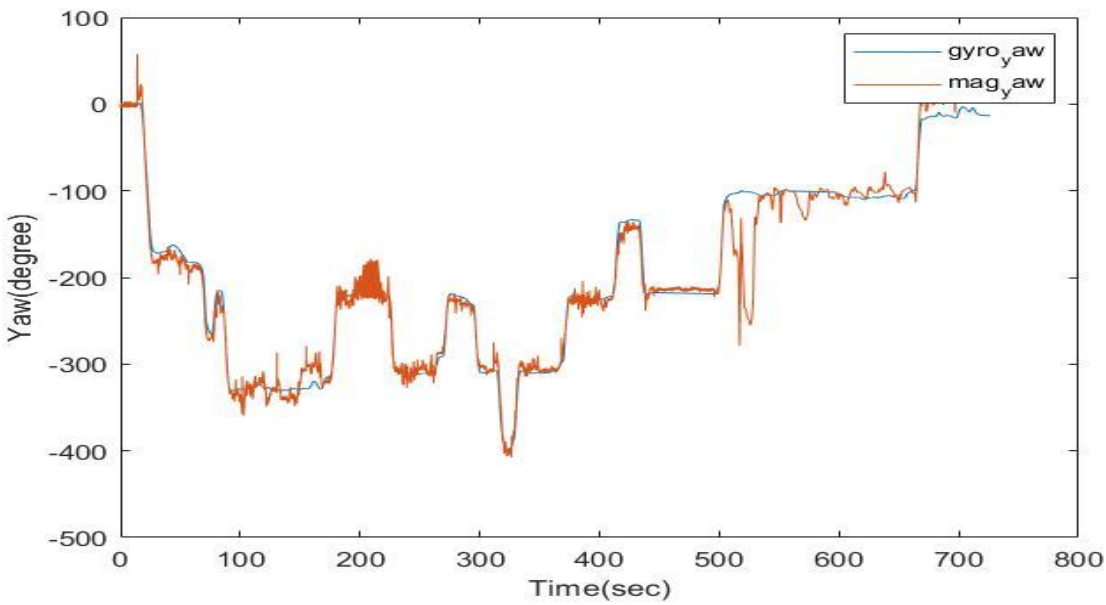
The corresponding yaw obtained from raw magnetometer data and calibrated magnetometer data is plotted and shown in the figure below.



The yaw from gyro is calculated by simply integrating the angular velocity(rad/sec) in Z thus obtaining yaw in radians. This yaw is then converted to degrees to get yaw from gyro in degrees.

Further all analysis is done on Calibrated Magnetometer data only.

Yaw calculated from gyro and calibrated magnetometer is shown in the figure below.



As the initial heading on which the integral is calculated in case of gyro is given as 0 the magnetometer yaw is also brought down to start from 0 and hence, we obtain the plot above and this is quite reasonable to do so when comparing yaw from gyro and magnetometer.

As the driving data is collected for a short span of 11min the drift in yaw calculated from gyro isn't too much but when calculating yaw from gyro using the

integral method for longer time periods the drift in yaw calculated from gyro increases unboundedly and hence this yaw has to be corrected with time from yaw calculated from other devices such as magnetometer.

So, a complimentary filter in this case will be a huge help to get correct yaw for longer time periods.

To implement complimentary filter first the Magnetometer data is filtered to remove high frequency noise using low pass filter and gyro data is filtered to remove low frequency noise using high pass filter.

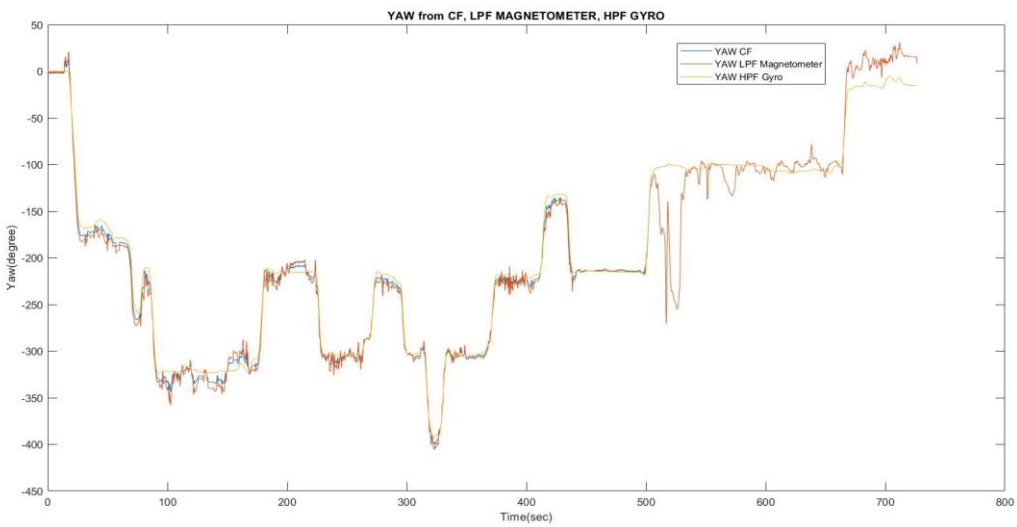
We have to filter the data, as just using unfiltered data in complimentary filter will cause wrong estimation of yaw as complimentary filter work is to just add the data from two different sources with a factor of weightage.

The low pass filter cut off for the calibrated magnetometer yaw is 0.0002. This cutoff frequency was reached for low pass filter after converting time domain magnetometer data to frequency domain using fft function in matlab and analyzing the signal to get an estimate of the frequency in which I got my magnetometer data and through manually trying different values of cutoff frequency within the frequency I got my magnetometer data I selected 0.0002, as the magnetometer yaw vs time curve smoothens that is unnecessary noise is removed from data.

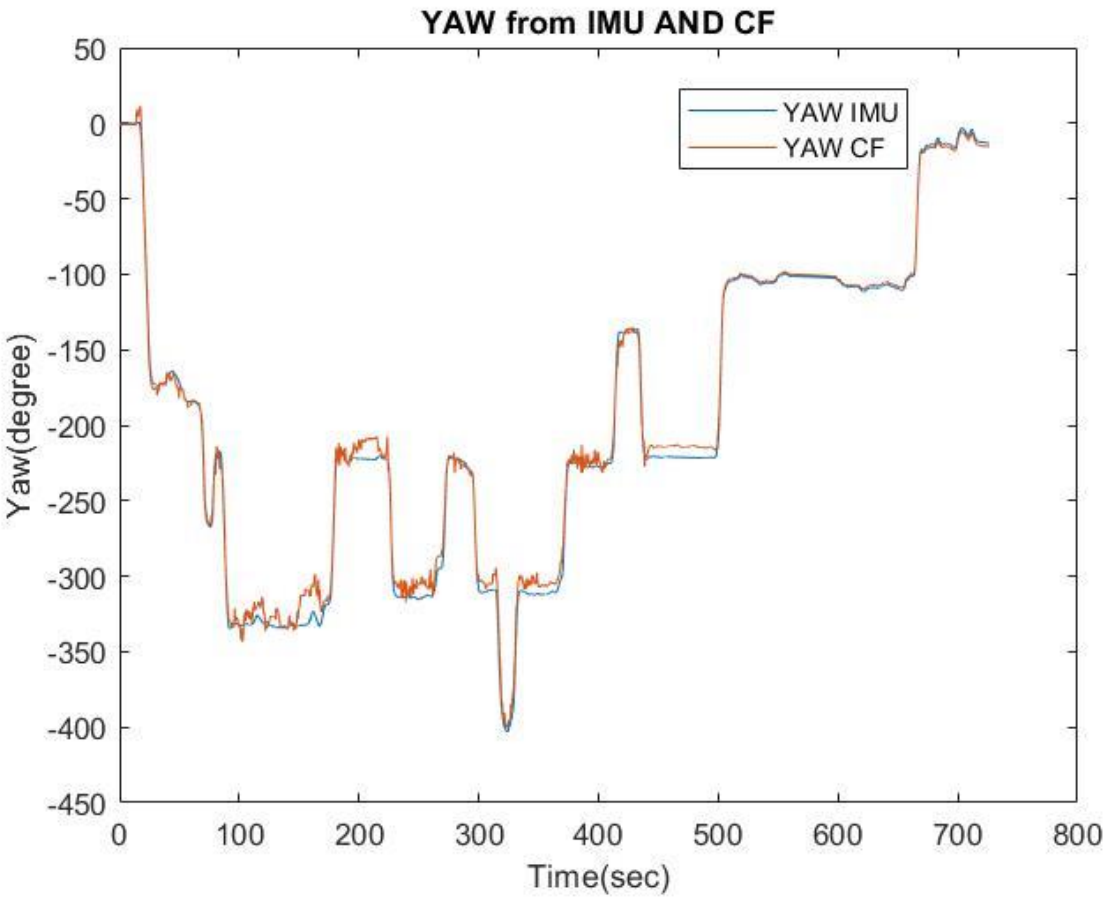
The gyro calculated yaw is passed through the high pass filter with cutoff frequency 0.00002. The selection of cutoff frequency for this high pass filter is also done using the same method as before.

The filtered yaw data is then passed through a complimentary filter which has a equation: a(Mag_Yaw)+(1-a)(Gyro_Yaw), here “a” I selected is 0.6

The figure below shows the Low pass Magnetometer calculated yaw, High pass Gyro calculated Yaw and Yaw from Complimentary Filter with Time.



The yaw obtained from Complimentary Filter is plotted against Yaw calculated by IMU in the figure below



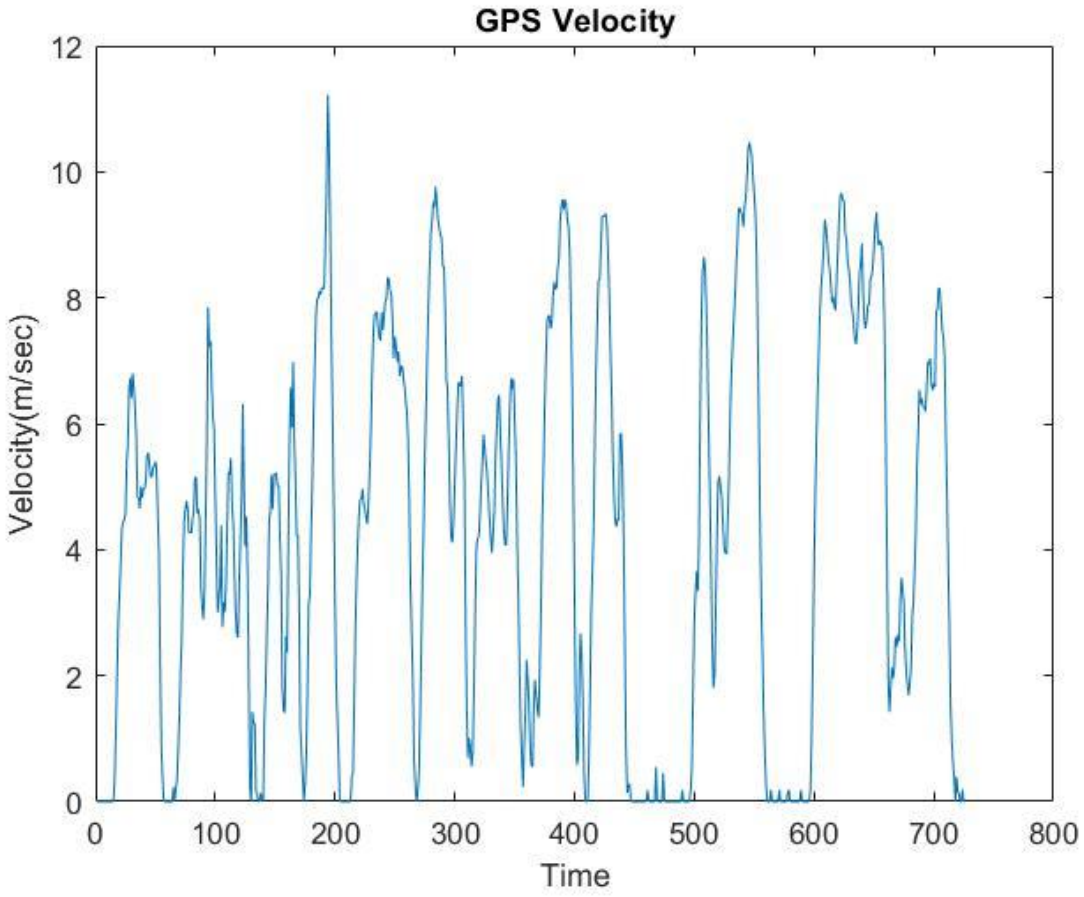
So, while navigation I would trust the yaw computed from the complimentary filter as the yaw computed from complimentary filter compensates for increase in bias from yaw calculated by gyro as this is the yaw, we got through integration which keeps on adding bias over time. The complimentary filter also does not completely eradicate the yaw calculated from gyro but takes a component of it based on tuning value to correct the yaw calculated from magnetometer as well.

So, this advantage of the complimentary filtered yaw makes it more reliable to me.

PART 2

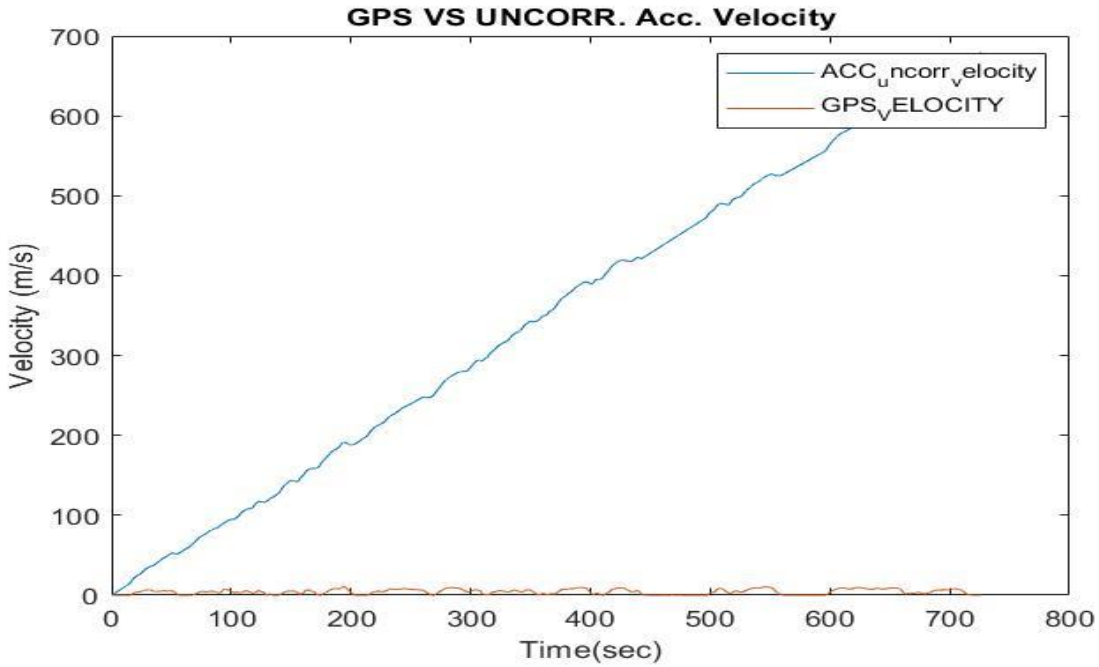
In Part2 we use the data from GPS and IMU while driving
The velocity from gps was calculated by using the formula
V= sqrt[(x2-x1)^2+(y2-y1)^2] / (1sec).
Where x2 and x1 are consecutive Easting Values and y2-y1 are consecutive Northing Values.
We divided by 1 as we want to get our velocity in m/s. Which is unnecessary though in our case as we have collected GPS data at a frequency of 1 only.

The figure below shows the velocity calculate using the GPS with respect to Time.



The velocity from accelerometer is calculated simply by integrating the acceleration obtained by accelerometer. In our case as our IMU x axis is aligned with the heading direction of the car so integrating the acceleration obtained in X direction will give us our heading velocity.

The figure below shows Velocity calculated from accelerometer with time along side with velocity calculated from GPS.



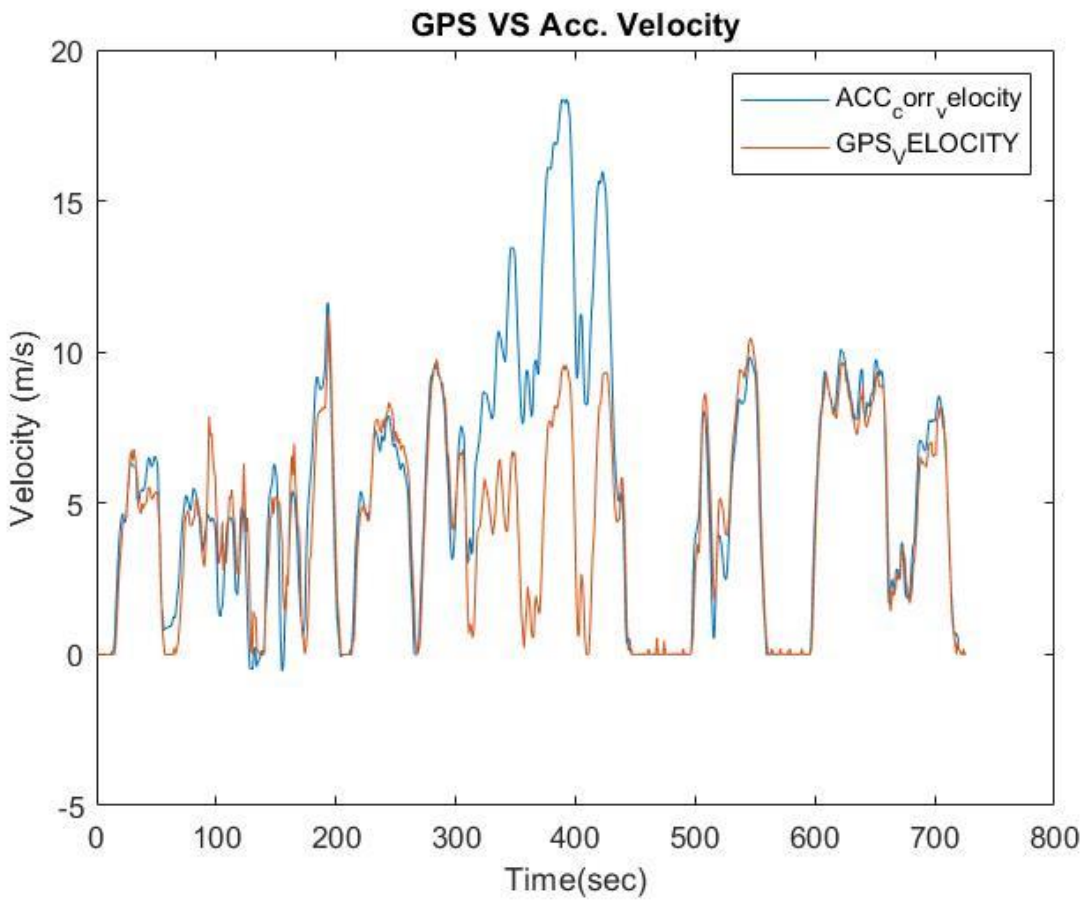
So clearly simply integrating the acceleration to obtain velocity is not justifiable.

The reason that our velocity keeps on increasing with time when calculated from accelerometer is that due to the pitch of our vehicle there is always a component of acceleration due to gravity in our heading direction which is added to the acceleration in X direction. This bias or error in acceleration when integrated over time keeps on increasing on each and every iteration hence our error in calculating velocity increases unboundedly.

So the component of acceleration due to gravity in our X direction can be removed in many ways, but the method that I used and sounded reasonable to me is to find out the jerk and then find the periods where our vehicle was at halt and as in these periods we exactly know that the vehicle acceleration in X direction is 0,we can simply take the mean of linear acceleration values in X direction in this period and subtract it from the linear acceleration in X data until we get to the next start of the vehicle halt period. This way of calculating acceleration due to gravity component in X also takes into account the change of gravity component in X due to change in pitch of the vehicle at each halt period there by reducing error in removing the gravity component in X direction.

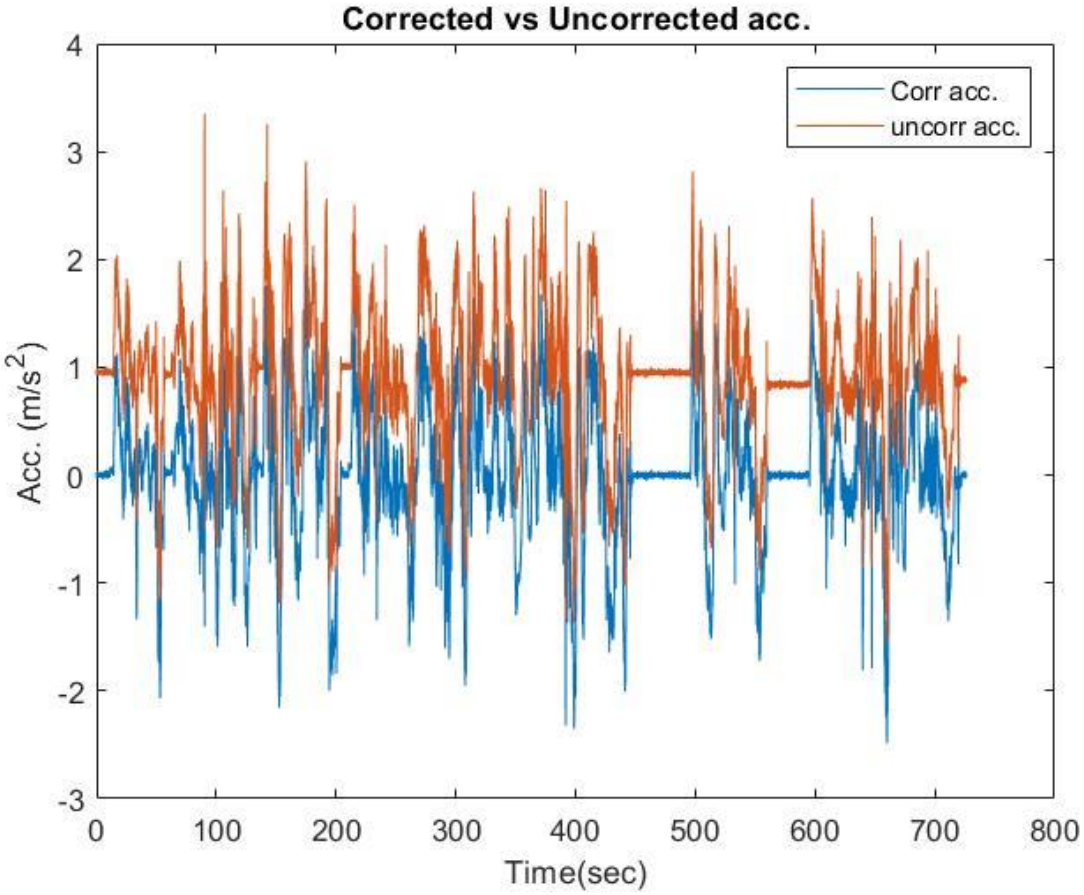
The program that I have written considers a vehicle to be at halt when the **magnitude of jerk is <0.04 consecutively for 50 points or more (> ~1sec).**

The corrected data of linear acceleration in X direction is now integrating to get the velocity and this is plotted with time alongside with GPS calculated velocity



The velocity calculated from corrected acceleration sounds more reasonable to me but not completely as the region between 300sec to 420sec overestimates the velocity of the vehicle. This is due to the fact of change in pitch of the vehicle in this region when compared to the halt position before the start of this region. This change in pitch of the vehicle in this region causes a change in component of acceleration due to gravity which is not accounted for by the calculation of mean in the previous halt region. Clearly looking at the graph one can say that the pitch of the vehicle increased when compared to the previous halt region pitch. Immediately after this velocity overestimate region there come a halt and the component of gravity in X direction is again calculated and hence the velocity calculated by accelerometer corrects itself after the velocity overestimated region.

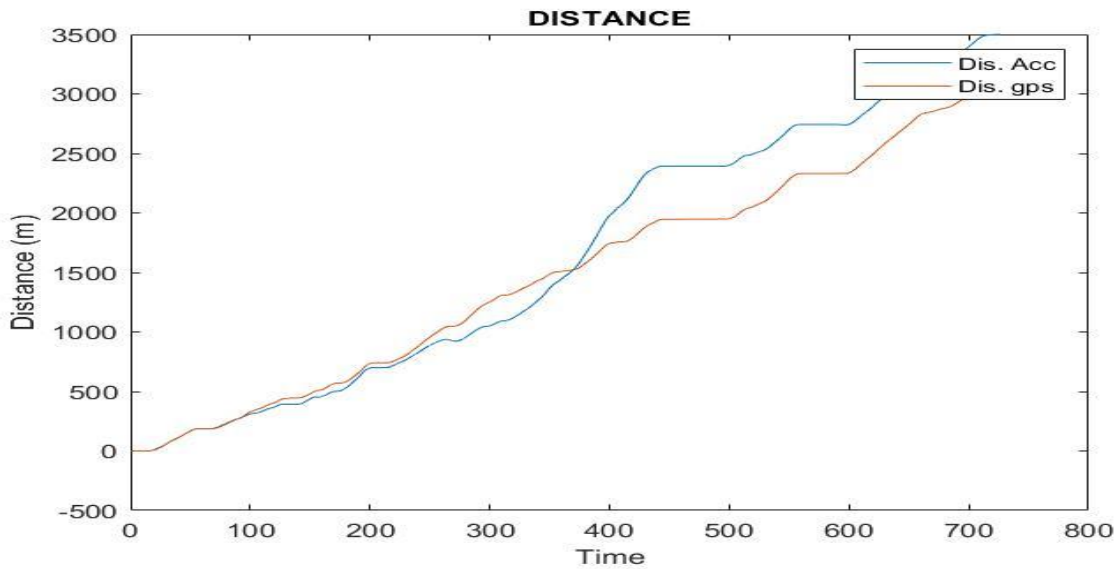
The figure below shows the Time vs Acceleration plot of Corrected and uncorrected acceleration.



PART 3

a.) Total Distance travelled calculated from GPS and Accelerometer

The total distance travelled is calculated by simply integrating the velocity with respect to time for both GPS and accelerometers

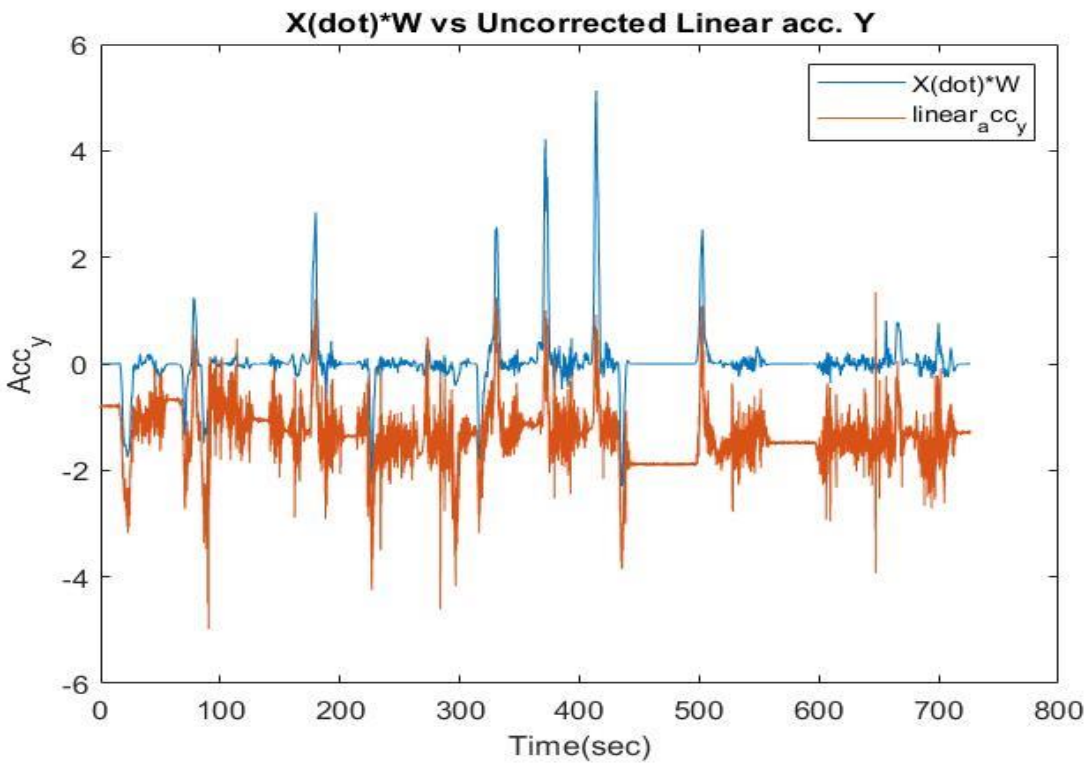


The distance travelled calculated from accelerometer is over estimated as the bias in the velocity calculated from accelerometer keeps on adding at each iteration of integration. Clearly from part 2 we have seen that the velocity is overestimated in a region between 300sec to 420sec and the error in this region alongside with minute error in velocity calculation in the other region causes a drift in the calculation of distance travelled using integration method in accelerometer.

b.) (Linear Velocity in X) *(Angular Velocity in Z) vs Linear Acceleration in Y

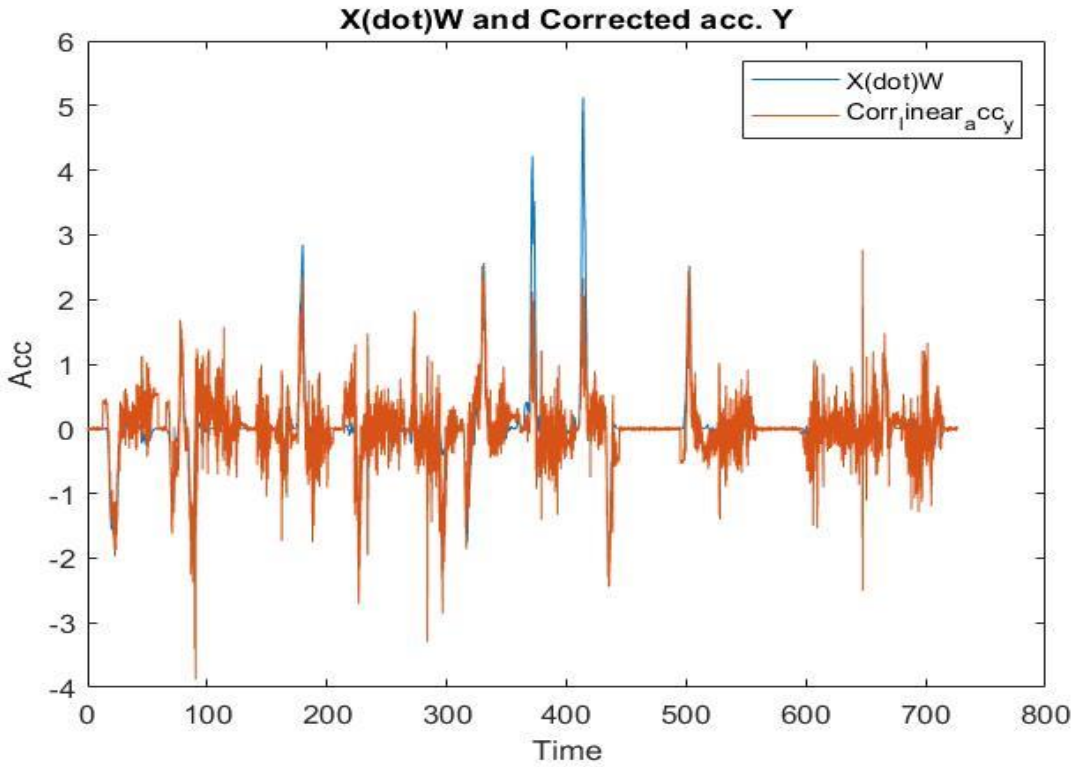
The product of linear velocity in X and angular velocity in Z gives a force acting in the direction perpendicular to the movement of vehicle in a plane. So, this is basically Coriolis force. In our case we can say that the product of linear velocity in X multiplied with Angular velocity in Z gives us Linear acceleration Y. As we are using the Linear Velocity in X that has been corrected for gravity related error this gives us Linear acceleration Y which is compensated for the component of gravity in Y direction due to roll of the car.

The time plot of (ẋ*ω) and Linear acceleration in Y from accelerometer is given below



One can clearly see from the figure above that linear acceleration in Y is offset below the zero due to component of gravity in the Y direction due to roll of the

car. At some points there is a sharp increase in value of linear acceleration in Y due to the turns that were made while driving and as while driving the roll of the car does not be constant due to steering and the plane on which the car is moving the component of gravity in the Y direction acceleration changes. The acceleration in Y can also be corrected using the same method as we corrected the linear acceleration in X, by doing so we plot the corrected linear acceleration in Y with (ẋ*ω).



Time(sec) vs Acceleration Y

By doing so we can reduce the component of gravity in Y. But at time period between 350sec to 420sec we can see a considerable variation in acceleration in Y given by (ẋ*ω) this is due to the fact that the velocity component in X was over predicted due to change in pitch of the car and gravity compensation on these points wasn't done as finely as required hence this variation is seen. As more is the value of velocity of X at these points the (ẋ*ω) overestimates the acceleration in Y direction and hence a peak variation is seen when this is compared to the gravity compensated Linear acceleration in Y calculated from accelerometer.

c.) Trajectory estimation from IMU and comparing with trajectory given by GPS

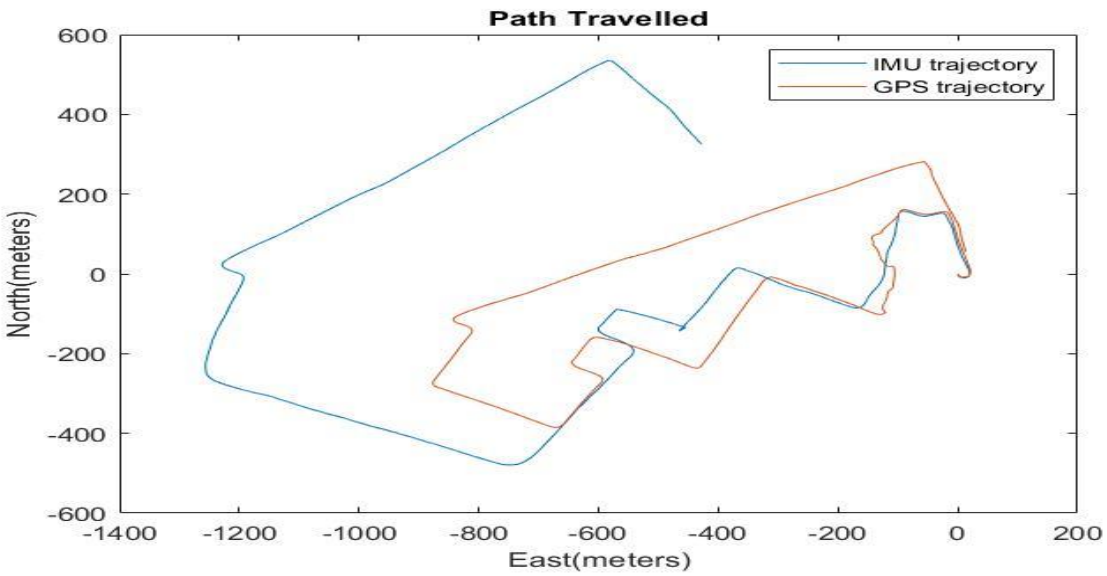
On plotting easting vs northing from GPS, we get the trajectory from GPS. The easting and northing are made to start from zero so as to compare the trajectory from GPS and IMU.

The trajectory from IMU is got by taking the component of corrected Linear velocity in X calculated from accelerometer in X and Y direction respectively using the yaw from complimentary filter. To get velocity component in X multiply the velocity calculated from accelerometer with cos(CF_yaw) and to get velocity component in Y multiply the velocity calculated from accelerometer with sin(CF_yaw).

Note:- The Complimentary yaw calculated was passed in the sin and cos function in radians.

The trajectory calculated from IMU was rotated by 180(degrees) about the X axis and rotation of ~57(degrees) was done about the Y axis to match the starting of the trajectory estimated from IMU with that of the GPS.

The trajectory from IMU and GPS is shown below



The trajectory estimated from IMU varies significantly after a certain time due to errors in compensating the component of gravity in our heading direction which

on integration causes rise in errors in estimation on velocity which on further integration and multiplication with yaw calculated from complimentary filter increases these errors by many times and hence the trajectory traced by IMU becomes unreliable for longer time durations. Moreover temperature fluctuations also effect the IMU sensor and Magnetometer is effected by magnetic disturbances around which makes the trajectory estimation by IMU more unreliable.